Weld-Inspection — NDT Procedure

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> **Audience:** Quality Worker (primary) · Maintenance Tech · Line Manager **Applies to:** Line-C, station 3, Weld-Inspection

1. Purpose

Weld-Inspection performs non-destructive testing (X-ray and ultrasonic) on every welded subassembly leaving station 2. This station has the most varied reject codes on Line-C and gates parts before Fastening at station 4. Reject disposition follows `Quality_Reject_Disposition_Policy.pdf`.

2. Normal operating envelope

- Ideal cycle time: **30 s**
- Max acceptable cycle time: **38 s**
- Expected reject rate: **≤ 4 %**
- Fault probability per cycle: **0.5 %**
- Input buffer capacity: **5 parts**

3. Idle reasons

`idle_reason`	Meaning here	First check
`Starved`	Welding-Robot (station 2) is not feeding parts.	Verify Welding-Robot state.
`Blocked`	Fastening-Station (station 4) cannot accept the inspected part.	Verify Fastening state.

4. Faults & corrective actions

4.1 `X-Ray-Tube-Fault`

- **Symptoms in telemetry:** `fault_type = "X-Ray-Tube-Fault"`. Image quality drops or imaging stops; reject rate may swing wildly.

- **Likely root cause:** X-ray tube reached end of life, or high-voltage arc-over.
- **Corrective action:**
  1. Lock out the X-ray cabinet per radiation safety.
  2. Verify HV power supply readings; replace the tube if filament is open.
  3. Run image quality indicator (IQI) test pieces.
- **Restart criteria:** IQI sensitivity meets the spec written on the recipe.

4.2 `Calibration`

- **Symptoms in telemetry:** `fault_type = "Calibration"`. Inspection reports either no defects ever or every part rejected.
- **Likely root cause:** automatic defect classifier drift; recipe parameters out of range.
- **Corrective action:**
  1. Run the calibration set (known-defect + known-good parts).
  2. Tune sensitivity until both pass and reject thresholds are met.
  3. Sign off recipe revision in the QMS.
- **Restart criteria:** 100 % accuracy on the calibration set.

4.3 `Film-Jam`

- **Symptoms in telemetry:** `fault_type = "Film-Jam"`. Inspection cycle cannot complete.
- **Likely root cause:** legacy radiographic media handler jammed (where applicable), or detector-panel transport stuck.
- **Corrective action:**
  1. LOTO. Clear the jam.
  2. Inspect transport rollers; replace worn ones.
  3. Verify panel positioning against the master target.
- **Restart criteria:** transport cycles 10 times with no slip.

5. Preventive maintenance schedule

- **Daily:** IQI verification at shift start.
- **Weekly:** detector clean; transport rollers inspected.
- **Monthly:** radiation safety integrity test.
- **Quarterly:** classifier retraining against the latest accepted defects.

6. References

- `Quality_Reject_Disposition_Policy.pdf`
- `Maintenance_Workflow_Overview.pdf`
- `Line-C_02_Welding-Robot_SOP.pdf`
- `Line-C_04_Fastening-Station_SOP.pdf`